

Synthetic invoice pilot — report example

From business context to a clear, reviewed direction.

Version 1 · Sep 11, 2026 · Business operations · English

Selected option: balanced

Report contents

2. Executive summary
3. Original request and underlying problem
4. Business context, objectives and success definition
5. Evidence register and confidence policy
6. Stakeholders and current system
7. Current-state workflow
8. Bottlenecks, gaps and root causes
9. Validation gaps and unanswered questions
10. AI necessity and automation opportunities
11. Three solution options
12. Option comparison and weighted decision matrix
13. Recommended option and rejected alternatives
14. Detailed scope and acceptance criteria
15. Technology stack and rationale
16. High-level architecture
17. Low-level component design
18. Future-state workflow and approval flows
19. User journeys, navigation and wireframes
20. Data model, ER diagram and database schema
21. REST API catalogue
22. Integration architecture
23. Security, privacy and governance
24. Infrastructure, cloud and deployment architecture
25. Human checkpoints and operating model
26. Implementation roadmap
27. Effort, cost and resource estimates
28. Timeline, milestones and sprint / release plan
29. Migration, training and adoption
30. Success KPIs and business value
31. Feasibility and readiness assessment
32. Risk register and mitigations
33. Independent Red Team review and design changes
34. Final recommendation and immediate next actions
35. Appendix: assumptions, unresolved items, glossary and artifact index

02 Executive summary

The bottleneck is duplicate entry, not reasoning complexity.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

03 Original request and underlying problem

Operations staff retype structured invoice rows; we need validated imports with human review.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Normalized problem: Duplicate entry between disconnected tools
 - Requested solution: Use AI for invoices
-

04 Business context, objectives and success definition

The operations team processes invoices.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Track duplicate entry count before and after pilot
-

05 Evidence register and confidence policy

Source-backed facts are attributed below. Proposed designs are recommendations, inferred assumptions are not confirmed facts, and estimates are planning ranges. Unknown facts, vendor access and applicable regulations require validation. Readiness scores are advisory heuristics.

Evidence / assumptions: User messages and uploaded documents. No independent verification is implied.

Source register

Category	Fact	Source
workflow	Staff retype invoice data	User message

06 Stakeholders and current system

Operations staff wants less duplicate entry; the reviewer needs a clear approval trail; the owner needs reconciled records.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Stakeholders and personas

Design aspect	Detail
Operations staff	Upload records, correct errors and view assigned imports.
Reviewer	Approve exports; cannot change payment permissions.

People, process, technology and data

Aspect	Current state
People	Operations staff
Process	Copy invoice details, review, then import.
Technology	Email; Spreadsheet
Data	Invoice records

07 Current - state workflow

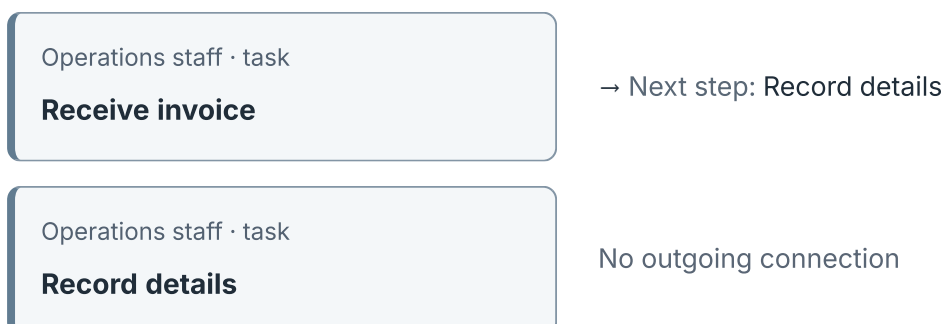
Copy invoice details, review, then import.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

Current activities

Step / owner	Activity / tools
1. Receive invoice — Operations staff	Read incoming email. Tools: Email
2. Record details — Operations staff	Retype invoice details. Tools: Spreadsheet

Current - state process map



08 Bottlenecks, gaps and root causes

Duplicate entry between disconnected tools

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

Duplicate entry

Analysis	Detail
Finding	Staff retype the same invoice data.
Evidence	User message: invoices copied from email to spreadsheet.
Confidence	90% advisory

Duplicate entry

Analysis	Detail
Finding	Staff retype the same invoice data.
Evidence	User message: invoices copied from email to spreadsheet.
Confidence	90% advisory

09 Validation gaps and unanswered questions

Resolve these items with the named stakeholders before committing to implementation.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- API access requires validation.
- Confirm accounting CSV format, retention period, baseline volume and rates.

10 AI necessity and automation opportunities

Automate validated imports with human exception review.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Decision: AUTOMATION_SUFFICIENT · AI necessity 15/100 · confidence 85%
- The supplied workflow follows fixed rules.
- Non - AI alternative: Validate and import structured invoice records using deterministic rules.

11 Three solution options

Compare scope, cost, delivery and operating trade-offs before selecting the implementation path.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

Lean: Lean process pilot

Comparison field	Option detail
Problem coverage / scope	Standardize spreadsheet entry and review.
Included	Standardize spreadsheet entry and review.
Excluded	Automated payments AI extraction
AI usage	none: Fixed fields and explicit validation rules do not need AI.
Technology / implementation	Configure the existing spreadsheet and a review checklist.
Integrations / data	Accounting import format and credentials require validation.
Security / privacy / governance	Role-limited access and audit records; jurisdiction and retention require validation.
Team / ownership	Operations owner approves rules; delivery engineer implements the pilot.
Benefits / outcomes	Reduce duplicate entry and make exception handling visible.
Limitations / trade-offs	Manual retyping remains
Risks / mitigations	Unverified integration: validate CSV format and reconcile a sample before rollout.
Scalability	One team
Best fit	Very limited budget
Do not select when	Volume makes manual entry impractical
Effort estimate	3–5 person-days. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.

Comparison field	Option detail
Cost estimate	Unknown — requires validation. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: unknown. Assumptions: Confirm scope, staff availability and rates with the operations owner.
Delivery range	1–2 weeks. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.

Balanced: Balanced import workflow

Comparison field	Option detail
Problem coverage / scope	Add a validated invoice import service with human review.
Included	Add a validated invoice import service with human review.
Excluded	Automated payments AI extraction
AI usage	none: Fixed fields and explicit validation rules do not need AI.
Technology / implementation	Small deterministic importer with a relational audit store.
Integrations / data	Accounting import format and credentials require validation.
Security / privacy / governance	Role-limited access and audit records; jurisdiction and retention require validation.
Team / ownership	Operations owner approves rules; delivery engineer implements the pilot.
Benefits / outcomes	Reduce duplicate entry and make exception handling visible.
Limitations / trade-offs	Requires a maintained import service
Risks / mitigations	Unverified integration: validate CSV format and reconcile a sample before rollout.
Scalability	Several teams with bounded imports

Comparison field	Option detail
Best fit	Repeatable invoice workload
Do not select when	No engineering owner is available
Effort estimate	10–20 person -days. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.
Cost estimate	Unknown — requires validation. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: unknown. Assumptions: Confirm scope, staff availability and rates with the operations owner.
Delivery range	3–6 weeks. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.

Advanced: Advanced integration platform

Comparison field	Option detail
Problem coverage / scope	Introduce a multi -team integration platform and stronger governance.
Included	Introduce a multi -team integration platform and stronger governance.
Excluded	Automated payments AI extraction
AI usage	none: Fixed fields and explicit validation rules do not need AI.
Technology / implementation	Central integration runtime with separate work queues and managed identity.
Integrations / data	Accounting import format and credentials require validation.
Security / privacy / governance	Role -limited access and audit records; jurisdiction and retention require validation.
Team / ownership	Operations owner approves rules; delivery engineer implements the pilot.

Comparison field	Option detail
Benefits / outcomes	Reduce duplicate entry and make exception handling visible.
Limitations / trade-offs	Higher operational burden
Risks / mitigations	Unverified integration: validate CSV format and reconcile a sample before rollout.
Scalability	Multi-team governance
Best fit	Many systems and large transaction volumes
Do not select when	Current scale cannot justify additional infrastructure
Effort estimate	30–60 person-days. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.
Cost estimate	Unknown — requires validation. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: unknown. Assumptions: Confirm scope, staff availability and rates with the operations owner.
Delivery range	8–14 weeks. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.

12 Option comparison and weighted decision matrix

Weights total 100%. Scores run from 1 (weakest fit) to 5 (best fit). Total = $\text{sum}(\text{weight} \times \text{score}) \div 5$, calculated by shift.AI. Scores and assumed priorities require stakeholder validation.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

Weighted scores

Criterion / weight	Lean	Balanced	Advanced
Business fit (40%)	2	5	4
Cost fit (30%)	5	4	1
Time to value (30%)	5	4	2

Criterion / weight	Lean	Balanced	Advanced
Total / 100	76.0	88.0	50.0

Business fit

Decision basis	Explanation
Weight basis	Assumed priority: confirm with operations owner.
Lean	2/5 — Advisory fit for a small operations team.
Balanced	5/5 — Advisory fit for a small operations team.
Advanced	4/5 — Advisory fit for a small operations team.

Cost fit

Decision basis	Explanation
Weight basis	Assumed priority: confirm with operations owner.
Lean	5/5 — Advisory fit for a small operations team.
Balanced	4/5 — Advisory fit for a small operations team.
Advanced	1/5 — Advisory fit for a small operations team.

Time to value

Decision basis	Explanation
Weight basis	Assumed priority: confirm with operations owner.
Lean	5/5 — Advisory fit for a small operations team.
Balanced	4/5 — Advisory fit for a small operations team.
Advanced	2/5 — Advisory fit for a small operations team.

13 Recommended option and rejected alternatives

Balanced: Balances removal of duplicate entry with manageable delivery effort.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- A simpler invoice workflow: Replace duplicate entry with reviewed imports.
- Solution type: CUSTOM_AUTOMATION · Complexity: LOW

- Lean rejected: Retains manual retyping.
- Advanced rejected: Scale and governance burden are not justified.
- Decision sensitivity: Choose lean if no engineer can maintain the importer.

14 Detailed scope and acceptance criteria

Import validated invoice rows, detect duplicates and require review before accounting export.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Detailed scope and acceptance criteria

Design aspect	Detail
REQ-1	Reject duplicate supplier/reference pairs; preserve a reason for every rejected row.
Out of scope	Do not automate payments or infer invoice fields using AI.

15 Technology stack and rationale

Use deterministic validation and a small persistent audit trail; verify existing hosting before choosing a vendor.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Technology stack and rationale

Design aspect	Detail
Backend	Python service: simple CSV validation; alternative is existing accounting import configuration.
Data store	Relational audit store; use the current supported database if available.
Identity	Reuse the organization identity provider if accessible; do not assume it exists.

16 High-level architecture

Operations supplies structured invoices to the Invoice importer, reviews exceptions, then approves the accounting handoff.

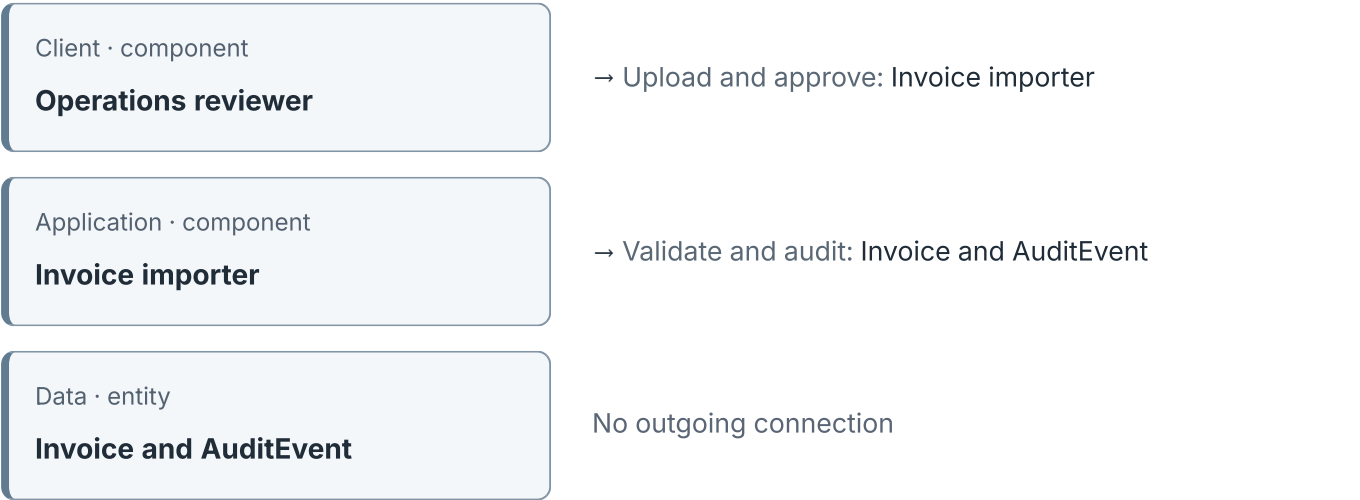
Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

- Data flow: Invoice arrives
- Data flow: Validate fields
- Data flow: Human approves
- Data flow: Import accounting record

Reviewed solution components

Component	Responsibility / rationale
Invoice importer	Validate and import structured records. Why: Fields follow deterministic rules. (Non - AI component)

High-level architecture



17 Low-level component design

Invoice importer owns parsing, field checks, deduplication and export audit. Human approval precedes external changes.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Low-level component design

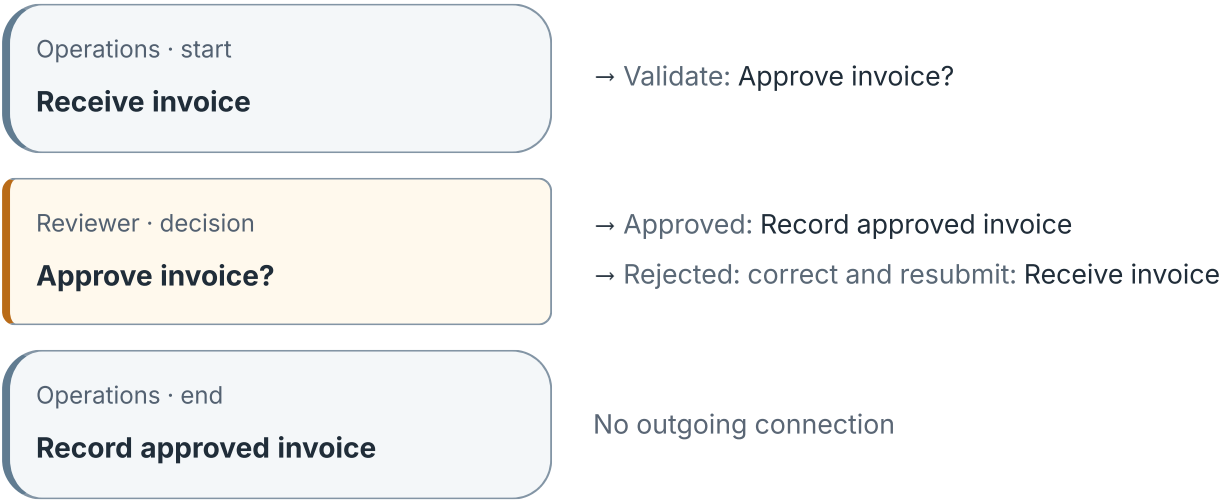
Design aspect	Detail
Input / output	CSV invoice rows → accepted rows, exception report and approved export.
Business rule	Unique supplier/reference; required amount and date; reject malformed rows.
Failure / security	Reject malformed imports atomically, audit attempt, require reviewer permission.

18 Future - state workflow and approval flows

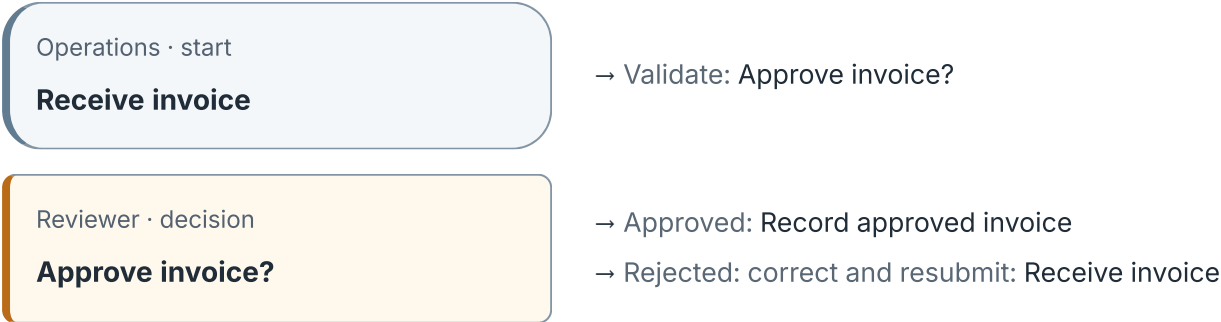
Upload records, validate fields and duplicates, resolve exceptions, review and approve export, then reconcile the result.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Future - state workflow and approval flows



Future - state workflow and approval flows



Operations · end

Record approved invoice

No outgoing connection

19 User journeys, navigation and wireframes

User journeys and navigation: An operations user starts in the import queue, uploads a CSV, reviews exceptions and hands approved records to the reviewer. Failure returns the user to correction with row-specific guidance.

Key screen concepts: Use an import queue, exception review and audit history. All views need empty, loading, success and error states with accessible labels.

Evidence / assumptions: User journeys and navigation: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.
Key screen concepts: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

User journeys and navigation

Design aspect	Detail
Goal / end state	Approved export is reconciled with the source invoice count.
Permissions / exceptions	Only reviewers approve; invalid files stay in correction and produce no export.

User journeys and navigation

Operations · start

Receive invoice

→ Validate: Approve invoice?

Reviewer · decision

Approve invoice?

→ Approved: Record approved invoice
→ Rejected: correct and resubmit: Receive invoice

Operations · end

Record approved invoice

No outgoing connection

Import queue

Operations staff · Start an import and track progress.

navigation

Queue / New import / History

Navigate between upload, exceptions and audit history.

input

Invoice CSV

Choose a structured CSV; explain required columns and maximum file size.

button

Validate file

Disabled while validating; show an actionable error for malformed files.

table

Recent imports

File, submitted by, status, accepted and rejected rows. Empty: no imports yet.

Review exceptions

Operations reviewer · Resolve errors before approving export.

table

Invoice rows

Reference, amount, validation result and reason; highlight duplicate or missing fields.

input

Corrected value

Show validation feedback next to the field; retain the original value in the audit trail.

button

Approve export

Only reviewers can approve; block unresolved errors and show a confirmation summary.

Import history

Operations owner · Reconcile approved exports and inspect audit events.

select

Date and status

Filter completed, failed or pending imports.

table

Audit events

Invoice reference, actor, action, time and export identifier. Retry failures without duplicate export.

metric

Reconciliation

Compare accepted source rows with exported rows; success requires matching counts and references.

20 Data model, ER diagram and database schema

Data model and ER diagram: Invoice records have an immutable ID; each invoice can have many audit events. Keep the accounting record as the financial source of truth.

Database schema: Invoice stores validated business fields; audit events record who approved or rejected a row. Retention must be agreed before production.

Evidence / assumptions: Data model and ER diagram: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

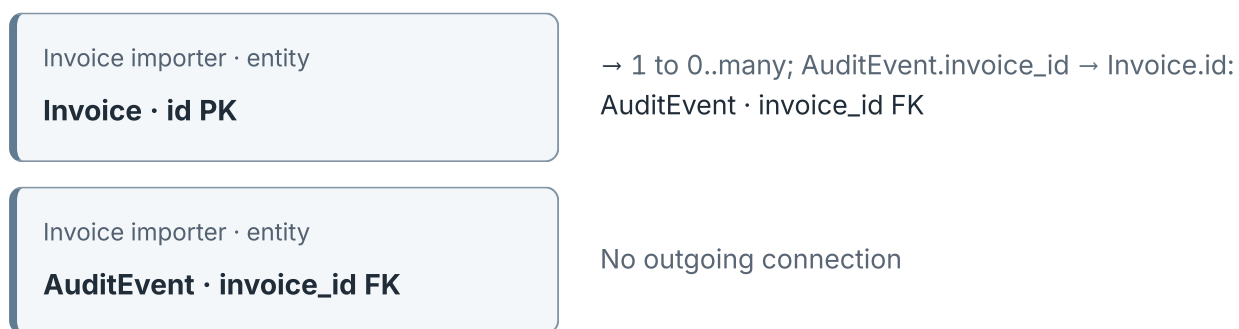
Database schema: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

- Structured invoice records

Database schema

Design aspect	Detail
Invoice	id UUID PK; supplier_ref TEXT; invoice_ref TEXT; amount DECIMAL; status TEXT; created_at TIMESTAMP; unique supplier_ref + invoice_ref.
AuditEvent	id UUID PK; invoice_id UUID FK → Invoice.id; actor_id TEXT; event TEXT; created_at TIMESTAMP; index(invoice_id, created_at).

Data model and ER diagram



schema.sql

```

CREATE TABLE Invoice (
  id UUID PRIMARY KEY,
  supplier_ref TEXT NOT NULL,
  invoice_ref TEXT NOT NULL,
  amount DECIMAL(12,2) NOT NULL,
  status TEXT NOT NULL,
  created_at TIMESTAMP NOT NULL,
  UNIQUE (supplier_ref, invoice_ref)
);
CREATE TABLE AuditEvent (
  id UUID PRIMARY KEY,
  invoice_id UUID NOT NULL REFERENCES Invoice(id),
  actor_id TEXT NOT NULL,
  event TEXT NOT NULL,
  created_at TIMESTAMP NOT NULL
);
CREATE INDEX audit_invoice_time ON AuditEvent(invoice_id, created_at);

```

21 REST API catalogue

Proposed internal API for the pilot; validate the final contract with implementers. No vendor API capability is assumed.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

REST API catalogue

Design aspect	Detail
POST /imports	Reviewer - authenticated CSV upload; validates rows; returns import status and exception references. 400 malformed file, 403 access denied, 409 duplicate idempotency key.
GET /invoices	Operations role; paginated invoice status; response contains id, reference and status. 403 unauthorized, 503 store unavailable.

openapi.yaml

```
openapi: 3.0.3
info:
  title: Proposed invoice pilot API
  version: 1.0.0
security:
  - sessionCookie: []
paths:
  /invoices:
    get:
      summary: List invoice status for authorized operations staff
      parameters:
        - in: query
          name: page
          schema: {type: integer, minimum: 1}
      responses:
        '200': {description: Paginated invoice IDs, references and status}
        '403': {description: Operations role required}
        '503': {description: Audit store unavailable}
  /imports:
    post:
      summary: Validate CSV rows for an authorized reviewer
      parameters:
        - in: header
          name: Idempotency-Key
          required: true
          schema: {type: string}
      requestBody:
```

```

    required: true
    content:
      text/csv:
        schema: {type: string}
  responses:
    '200': {description: Import status and exception references}
    '400': {description: Malformed file or invalid fields}
    '403': {description: Reviewer role required}
    '409': {description: Idempotency key conflicts with another payload}
  components:
    securitySchemes:
      sessionCookie:
        type: apiKey
        in: cookie
        name: session

```

22 Integration architecture

Accounting CSV exchange is proposed pending format validation. Reconcile counts and references; never retry a completed export without checking its identifier.

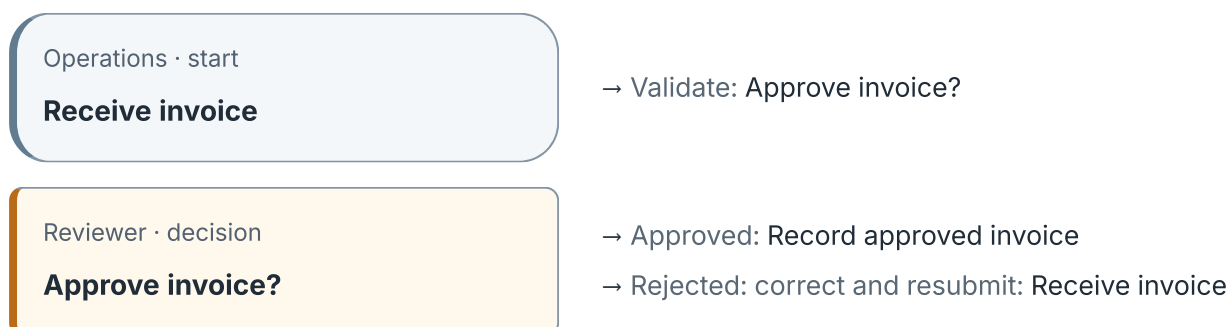
Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

- Accounting API access to be validated

Integration architecture

Design aspect	Detail
Mapping / auth	Map invoice reference and amount to the approved accounting template; only the reviewer exports.
Failure / source of truth	Reject format mismatches; preserve source rows and reconcile against accounting.

Integration architecture



23 Security, privacy and governance

Restrict invoice access to operations reviewers; record approvals and support access. Confirm retention, jurisdiction and accounting permissions before pilot.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Security, privacy and governance

Design aspect	Detail
Boundary	No payment credentials or automatic payment authorization.
Control	Encrypted transport, least-privilege access and auditable export approval.

24 Infrastructure, cloud and deployment architecture

Infrastructure and cloud architecture: Pilot in an isolated environment with bounded file storage, private database access and monitored job outcomes.

Deployment and release architecture: Validate in test, reconcile a representative sample, then promote an approved release. Operations owns go-live approval.

Evidence / assumptions: Infrastructure and cloud architecture: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Deployment and release architecture: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Infrastructure and cloud architecture

Design aspect	Detail
Backup / recovery	Agree recovery objectives, then test restoring the invoice audit trail.
Scale / secrets	Limit file size; keep credentials server-side in the approved secrets store.

Deployment and release architecture

Design aspect	Detail
CI/CD	Run schema and duplicate-handling checks before promotion.
Rollback	Stop new imports, retain audit data and return to reviewed manual entry.

25 Human checkpoints and operating model

Operations reviews exceptions and approves each export. The delivery engineer investigates failed jobs; the operations owner owns reconciliation.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

- Review exceptions and approve imports

Human checkpoints and operating model

Design aspect	Detail
Checkpoint	Do not export until exceptions are resolved or explicitly excluded.

26 Implementation roadmap

Replace duplicate entry with reviewed imports.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

01: Validate a small pilot

Actions	Ready when
Confirm import format Test duplicate handling	Pilot imports reconcile with source invoices

27 Effort, cost and resource estimates

Effort and cost estimates: Illustrative pilot: 10–20 person-days over 3–6 elapsed weeks, assuming one engineer and part-time operations review. Rates and vendor prices are unknown.

Resource plan: Assume one delivery engineer and a part-time operations reviewer; validate availability before scheduling.

Evidence / assumptions: Effort and cost estimates: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.
Resource plan: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

- Implementation effort: 10–20 person-days. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.
- Elapsed duration: 3–6 weeks. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: low. Assumptions: Confirm scope, staff availability and rates with the operations owner.
- Development/setup cost: Unknown — requires validation. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: unknown. Assumptions: Confirm scope, staff availability and rates with the operations owner.
- Recurring hosting/support: Unknown — requires validation. Basis: Illustrative planning assumption; no vendor quote or measured baseline supplied. Confidence: unknown. Assumptions: Confirm scope, staff availability and rates with the operations owner.

Effort and cost estimates

Design aspect	Detail
Validation	2–4 person-days (low confidence); confirm format and representative data.
Implementation / acceptance	8–16 person-days (low confidence); depends on access and reviewer availability.

Resource plan

Design aspect	Detail
Engineer	Implement validation, audit and deployment; 10–20 person-days, low-confidence assumption.
Operations owner	Confirm rules, sample reconciliation and go-live; involvement requires validation.

28 Timeline, milestones and sprint / release plan

Timeline and milestones: Tentative elapsed duration 3–6 weeks. External access and reviewer availability are critical dependencies, so no calendar dates are committed.

Sprint and release plan: Release by validation gate rather than unsupported fixed sprint dates.

Evidence / assumptions: Timeline and milestones: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.
Sprint and release plan: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Timeline and milestones

Design aspect	Detail
Week 1–2	Confirm data/format and pilot acceptance criteria.
Week 2–5	Implement and reconcile sample imports after format approval.
Week 3–6	Pilot, reconcile exports and approve go-live only after failure/rollback checks.

Sprint and release plan

Design aspect	Detail
MVP	CSV validation, duplicate checks and exception review.
Pilot	Approved export, audit history and sample reconciliation.
Rollout	Expand only after operations accepts results and rollback is exercised.

29 Migration, training and adoption

Train staff using sample invoices and exception cases. Reconcile any migrated open items, retain the old spreadsheet for an agreed period and review feedback weekly.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Migration, training and adoption

Design aspect	Detail
Measure adoption	Count completed imports, exceptions and manual re-entry before and after the pilot.

30 Success KPIs and business value

Less duplicate entry; savings must be measured.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Track duplicate entry count before and after pilot

Business value estimates

Metric	Estimate / basis
Manual entry	Potential reduction, not yet quantified — No measured time baseline was supplied. (Assumption / estimate)

31 Feasibility and readiness assessment

Readiness remains advisory; the integration and baseline workload have not been validated.

Evidence / assumptions: Synthetic fixture: proposed design based on manual invoice entry; integration, rates and dates require validation.

Readiness assessment

Design aspect	Detail
Digital maturity	Developing — current spreadsheet workflow, no measured maturity baseline.
AI readiness	Unknown; no AI required for the selected scope.
Automation opportunity	Deterministic field validation and deduplication.
Project health	Planning stage; timeline assumptions pending confirmation.
Implementation readiness	Blocked on import format and representative data.
Solution quality	Requires reconciliation and failure/rollback testing.

Feasibility (advisory)

Dimension	Score / reason
Technical	75/100 — Validate access in a small pilot.
Data	75/100 — Validate access in a small pilot.
Integration	75/100 — Validate access in a small pilot.

Dimension	Score / reason
Operational	75/100 — Validate access in a small pilot.
Business	75/100 — Validate access in a small pilot.
Overall	75/100

32 Risk register and mitigations

Mitigations are proposed controls; residual risks and validation actions remain visible.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

Accounting import format unavailable

Risk field	Assessment
Severity	high
Likelihood	unknown
Impact	Blocks validated export.
Mitigation	Confirm a representative CSV import before implementing the handoff.
Owner	Operations owner
Validation Action	Reconcile a sample with accounting.
Residual Concern	Vendor changes may require remapping.

Accounting API access is not confirmed

Review field	Assessment
Category / severity	INTEGRATION / HIGH
Finding / impact	The supplied workflow names an accounting tool but no available API.
Mitigation	Validate API access before implementation; use a reviewed CSV import as fallback.
Status	Mitigation proposed; validate before implementation

33 Independent Red Team review and design changes

Confirm integration access before building.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Proposal 1: Replace duplicate entry with reviewed imports. Components: Invoice importer: Validate and import structured records. Human controls: Review exceptions and approve imports
Integrations: Accounting API access to be validated

Red Team cycle 1

Summary	Review scope
Confirm integration access before building.	Selected solution and all implementation design parts

Accounting API access is not confirmed

Review field	Assessment
Category / severity	INTEGRATION / HIGH
Finding / impact	The supplied workflow names an accounting tool but no available API.
Mitigation	Validate API access before implementation; use a reviewed CSV import as fallback.
Status	Mitigation proposed; validate before implementation

34 Final recommendation and immediate next actions

Pilot deterministic imports with staff review.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Confirm integration access
- Measure the current workload

35 Appendix: assumptions, unresolved items, glossary and artifact index

Validate these assumptions and unresolved items before implementation. Design artifacts are proposals for review.

Evidence / assumptions: Proposed design / advisory analysis; validate against the evidence register.

- Do not automate payments
- CSV import is supported; validate before implementation
- API access requires validation.
- Confirm accounting CSV format, retention period, baseline volume and rates.
- Idempotency: repeating a request does not duplicate its effect.

Generated artifact index

Section	Artifact
Current - state workflow	Current - state process map
High - level architecture	High - level architecture
Future - state workflow and approval flows	Future - state workflow and approval flows
Future - state workflow and approval flows	Future - state workflow and approval flows
User journeys, navigation and wireframes	User journeys and navigation
User journeys, navigation and wireframes	Import queue
User journeys, navigation and wireframes	Review exceptions
User journeys, navigation and wireframes	Import history
Data model, ER diagram and database schema	Data model and ER diagram
Data model, ER diagram and database schema	schema.sql
REST API catalogue	openapi.yaml
Integration architecture	Integration architecture

Prepared with shift.AI. Recommendations, estimates and readiness scores are advisory. Validate assumptions and unresolved risks before implementation.